

AI Fashion Design: Project Proposal

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AI Powered Virtual Fashion Assistant

Project Idea

The purpose of this project is to develop an AI-powered fashion recommendation system that provides personalized clothing and color suggestions based on a user's skin tone. The system will use computer vision techniques to analyze a user's uploaded facial image, determine their skin tone category, and recommend suitable color palettes and fashion styles.

Problem Statement

Choosing clothing that complements an individual's appearance can be challenging. Many people are unsure which colors, styles, and fashion trends suit their skin tone and personal aesthetic. Existing fashion recommendation platforms often provide generic suggestions without considering individual characteristics.

There is a need for a personalized fashion assistance system that can analyze a user's facial image, identify their skin tone, and provide customized fashion recommendations. By using Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision techniques, users can receive personalized color palettes, outfit suggestions, and fashion moodboards that better match their appearance and style preferences.

Project Description

Many people struggle to identify clothing colors and styles that complement their appearance. This project aims to solve this problem by creating an intelligent styling assistant that offers personalized fashion recommendations.

The user will upload a selfie through the application. The system will detect the face, analyze skin tone characteristics, and classify the user into categories such as Fair, Medium, Olive, or Deep skin tones. Based on the detected skin tone, the system will recommend suitable color palettes and fashion aesthetics.

In addition, the application will generate a personalized fashion moodboard containing outfit inspirations, color combinations, accessories, and styling ideas that match the user's appearance and recommended fashion style.

System Functionality

The AI Powered Virtual Fashion Assistant will:

- Accept a facial image from the user.
- Detect the face using Computer Vision.
- Analyze the skin tone from the image.
- Classify the skin tone category.
- Predict a suitable seasonal color palette.
- Recommend clothing colors and fashion styles.
- Generate a personalized fashion moodboard.
- Display results through a user-friendly interface.

Example

Input	Output
User Selfie	Skin Tone: Medium
User Selfie	Season: Autumn
User Selfie	Recommended Colors: Brown, Olive Green, Beige
User Selfie	Personalized Fashion Moodboard

Objectives

The main objectives of this project are:

- To learn the fundamentals of Artificial Intelligence and Machine Learning.
- To understand Computer Vision and facial image analysis.
- To develop a skin tone detection system.
- To apply Machine Learning techniques in fashion recommendations.
- To generate personalized fashion moodboards.
- To improve software development and teamwork skills.
- To gain experience in integrating AI models into applications.

Tools and Technologies

Software

- Python
- Google Colab
- VS Code

- Streamlit

Technologies

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Computer Vision
- OpenCV
- K-Means Clustering
- Recommendation Systems
- Dataset

The project will utilize:

- User-uploaded facial images
- Sample skin tone datasets
- Fashion recommendation datasets
- Fashion image collections for moodboard generation

Project Scope

The project will focus on:

Skin Tone Detection

The system will classify users into categories such as:

- Fair
- Light
- Medium
- Olive
- Deep

Seasonal Color Analysis

Based on the users preferences, the system will recommend color palettes for different seasons:

- Spring - soft, elegant, cool colors
- Summer - fresh, bright, warm colors
- Autumn - rich, earthy, warm colors
- Winter - sharp, high-contrast, cool colors

Fashion Recommendations

The system will provide:

- Suitable clothing colors
- Outfit suggestions
- Style recommendations

Moodboard Generation

The system will generate:

- Personalized fashion inspiration boards
- Color palette visualizations
- Style-based outfit references

The project is designed for educational purposes and demonstrates how AI and Computer Vision can be applied in personalized fashion assistance.

Week Wise Deliverables

Week 5

- Research AI, ML, and Computer Vision concepts.
- Study skin tone analysis techniques.
- Finalize project requirements and design.

- Set up the development environment.
- Collect and organize datasets.

Week 6

- Preprocess datasets.
- Implement face detection.
- Develop skin tone classification module.
- Build recommendation logic.
- Test and evaluate model performance.

Week 7

- Develop the Streamlit user interface.
- Integrate AI and Computer Vision modules.
- Implement moodboard generation.
- Perform testing and debugging.
- Prepare project documentation.
- Prepare final presentation and submission.

Work Division

Since our team consists of four members, responsibilities will be distributed according to different project components.

Team Member	Responsibilities
Simone Shree Pathak Baiju	Dataset collection, data preprocessing, machine learning model development, model training and evaluation

Niyati Niroula	User interface development, application integration, testing and debugging
Anuttar Bajracharya	Computer Vision implementation, face detection, skin tone extraction, image processing
Aryan Shrestha	Fashion recommendation engine, moodboard generation, documentation and presentation preparation

All team members will:

- Participate in project planning.
- Discuss project progress and improvements.
- Test the final application.
- Prepare the final report.
- Contribute to presentation preparation.
- Deliver the final presentation.

Learning Expectations

Through this project, we hope to learn:

- Basic concepts of Artificial Intelligence and Machine Learning.
- Computer Vision techniques and image processing.
- Face detection and skin tone analysis methods.
- Recommendation system development.
- User interface design and application integration.

- Data preprocessing and model evaluation.
- Team collaboration and project management skills.
- Real-world applications of AI in fashion and personalization.

Conclusion

The AI Powered Virtual Fashion Assistant is a beginner-friendly project that combines Artificial Intelligence, Machine Learning, Computer Vision, and Recommendation Systems. The system will analyze a user's facial image, identify their skin tone, recommend suitable color palettes and fashion styles, and generate personalized fashion moodboards.

This project will provide practical experience in modern AI technologies while demonstrating a real-world application of personalized fashion recommendation systems. It will strengthen our technical skills and serve as a foundation for future AI and Computer Vision projects.